Positive Phase Moments and Structured Curvature Bounds for a Negatively Pinched Kähler Threefold

A Source-Specific Ram Analysis of M688

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M688

October 8, 2026

Abstract

We analyze the pinned construction of a complete, contractible Kähler threefold with two-sided negative real sectional curvature and no bounded holomorphic map to $\mathbb{C}^3$ whose Jacobian is everywhere nonzero. The representing densities are actual positive functions, not finite positive-semidefinite moment tables. Their endpoint correction annihilates every positive holomorphic degree exactly, as already proved in the source. Our additions are an exact Fourier-tail estimate for the regularized logarithmic tests, a transfer to separated nongeometric degree sequences with controlled exponential growth, an all-fixed-order weighted phase derivative estimate, and a quantitative mixed-block coercivity inequality for the actual real-plane curvature expression. The latter yields an explicit conservative sufficient metric parameter, without discarding the indefinite mixed term. We retain the source's global construction as an explicitly identified dependency and distinguish analytic proofs from finite algebra tests. Contribution assessment: C3/5, source-specific spectral and quantitative curvature improvements are proved, while the global geometric construction and its input constants remain inherited dependencies.

1 Statement, scope, and source map

Let $B = \mathbb{B}^2 \subset \mathbb{C}^2$, $S = \partial B$, and let σ be probability surface measure on S. Write

$$\psi(z) = -\log(1 - |z|^2), \qquad I = \partial\bar{\partial}\psi.$$

The source [1] constructs a smooth real function ϕ on B and the Hartogs domain and potential

$$M_\phi = \{(z, w) \in B \times \mathbb{C} : e^{\phi(z)}|w|^2 < 1\}, \qquad K_\lambda = \lambda\psi - \log(1 - e^{\phi}|w|^2). \tag{1}$$

For one sufficiently large, fixed, finite λ, its real metric $g = 2\,\mathrm{Re}(\partial\bar{\partial}K_\lambda)$ is complete and satisfies $-B_0 \le K_g(\Pi) \le -A_0 < 0$ on *every real two-plane*. No bounded holomorphic $F : M_\phi \to \mathbb{C}^3$ can have a Jacobian nonzero everywhere. This does not say that every such Jacobian is identically zero, or that all bounded holomorphic functions are constant: the base coordinates are bounded nonconstant functions.

The full twelve-file build was read: main file, bibliography, and every mathematical section. The following is the dependency map, not a claim of formal verification of all arguments.

Source files	Mathematical responsibility
01, 02	Headline theorem; base potential and holomorphic-minorant reduction.
03	Projective packing, homogeneous peaks, compact radial profiles.
04 construction	Exact phase corrections and all-degree endpoint annihilation.
04 estimates	Orbit doubling, weighted second derivative, noncircular positivity.
05, 06	Negative logarithmic tests, later-frequency interference, infinite potential, centered derivative bounds.
07	Smooth metric across the zero section, contractibility, completeness.
08	Full curvature tensor, inverse-index convention, real-plane Gram identities.
09	Degenerate-plane argument, uniform large-parameter sign, fixed-parameter two-sided pinching.

The new results below are identified individually. The source's packing construction, profile existence, and geometric transfer are credited to the source. No historical-priority claim is made. Applicability L4 describes relevance of the private methods, not the realized C3 assessment.

2 Exact positive densities: the inherited mechanism

For a sufficiently separated maximal family of projective directions $p \in S$, the source uses

$$P_k(z) = \sum_p \langle z, p\rangle^k, \qquad |P_k(z)| \le C_P|z|^k \quad (z \in \mathbb{C}^2).$$

The number of directions is between k/D_0^2 and $4k/D_0^2$. The global homogeneous bound, including outside the ball, is important for the anisotropic Cauchy estimates below. On a patch around p use

$$y = -k\log|\xi_1|, \quad \theta = \arg\xi_1, \quad \gamma = \arg\xi_2, \qquad \mathrm{d}\sigma = \frac{2}{k}e^{-2y/k}\,\mathrm{d}y\,\frac{\mathrm{d}\theta}{2\pi}\,\frac{\mathrm{d}\gamma}{2\pi}.$$

Patches $0 \le y < R_0$ are disjoint for the source's choices of D_0, k. There P_k differs from $e^{-y}e^{ik\theta}$ by a uniformly small packing tail. Fix the source's smooth real profiles

$$b' - b = f, \quad b(0) = b(R_0) = 0, \quad \int_0^{R_0} e^{-y}f(y)\,\mathrm{d}y = 0, \quad \|f\|_\infty = F < 1. \tag{2}$$

The profiles vanish appropriately near the outer endpoint; smoothness at the patch center is checked in (θ, ξ_2) coordinates.

Put $W_0 = 1$, $\ell_j = \sum_{i\le j} k_i$, and $s_p(\theta) = W_{j-1}(e^{i\theta}p)$. The correction is

$$A_{j,p} = e^{2y/k_j}\,\mathrm{Re}\left\{e^{ik_j\theta}\left(f(y)s_p(\theta) - \frac{b(y)}{ik_j}s_p'(\theta)\right)\right\}, \qquad W_j = W_{j-1} + \sum_p A_{j,p}. \tag{3}$$

The derivative of s_p is essential, not an optional small residual. If $s_p = \sum_{|s|\le\ell_{j-1}} \widehat{s}_p(s)e^{is\theta}$, the radial coefficient in positive frequency $m = k_j + s$ is $b' - (m/k_j)b$. For a holomorphic monomial $\xi_1^n\xi_2^l$, $l > 0$ is killed by the γ average. For $l = 0$, only the matching frequency can survive the θ average, and its radial integral is

$$\int_0^{R_0} e^{-ny/k_j}\left(b' - \frac{n}{k_j}b\right)\mathrm{d}y = \left[e^{-ny/k_j}b(y)\right]_0^{R_0} = 0. \tag{4}$$

When $k_j > \ell_{j-1}$ there is no zero frequency. Consequently

$$\int_S W_j\,\mathrm{d}\sigma = 1, \qquad \int_S \xi^\alpha W_j\,\mathrm{d}\sigma = 0 \quad (|\alpha| \ge 1).$$

These identities cover all degrees, not a finite moment fit. For holomorphic H on B and $0 \le r < 1$, they give

$$\int_S H(r\xi) W_j \, \mathrm{d}\sigma = H(0).$$

This follows by uniformly convergent Taylor expansion on the compact sphere of radius r.

For $k_j = Q^j$ the source proves, with a single sufficiently large Q,

$$c_0 W_{j-1} \le W_j \le 2W_{j-1}, \qquad |ZW_j| \le K_* \sqrt{k_j} W_j, \qquad |TW_j| \le K_* k_j W_j, \tag{5}$$

where T is common-phase differentiation and Z is a unit complex horizontal tangent. There is no positive lower bound uniform in j. The proof first fixes $c_0 = (1-F)/4$, then a normalized patch derivative constant C_2 independent of K_*, Q, then $K_* > 2C_2/c_0$. Orbit doubling controls the second derivative before Q is enlarged. Only afterwards does it impose the small ratios K_*/Q, C_1/Q^2, and $K_* D_R/\sqrt{Q}$. This order prevents circular selection of the positivity constants. We do not claim it as a new result.

3 Exact spectrum of the logarithmic test

The source fixes $\varepsilon > 0$, $1 < a < 2$, and uses

$$U_k(z) = F_\varepsilon(P_k(z)), \qquad F_\varepsilon(v) = \log \frac{|1 - av|^2 + \varepsilon^2}{1 + \varepsilon^2}.$$

It proves a negative average at stage j, uniformly when $1 - \eta \le r^{k_j} \le 1$: $\int U_{k_j}(r\xi) W_j \, \mathrm{d}\sigma \le -d_2$, where $d_2 > 0$. The next lemma sharpens the subsequent interference estimate; it does not replace that negative-average construction.

Lemma 3.1 (Exact phase coefficients)**.** *For $A \in \mathbb{C}$, put $u = |A|$, $b_0 = 1 + u^2 + \varepsilon^2$, and*

$$C_A = \frac{b_0 + \sqrt{b_0^2 - 4u^2}}{2}, \qquad \zeta_A = A/C_A, \qquad \rho_\varepsilon = \sqrt{1+\varepsilon^2} - \varepsilon = e^{-\operatorname{arsinh} \varepsilon}.$$

Then $|\zeta_A| \le \rho_\varepsilon < 1$, and the absolutely uniformly convergent phase expansion is

$$\log \frac{|1 - Ae^{i\theta}|^2 + \varepsilon^2}{1 + \varepsilon^2} = \log \frac{C_A}{1 + \varepsilon^2} - \sum_{m \ge 1} \frac{\zeta_A^m e^{im\theta} + \overline{\zeta_A}^m e^{-im\theta}}{m}. \tag{6}$$

Discarding all $|m| \ge M$, $M \ge 1$, has uniform error at most

$$E_M = \frac{2\rho_\varepsilon^M}{M(1 - \rho_\varepsilon)}. \tag{7}$$

Proof. The discriminant is positive since $b_0 - 2u = (u-1)^2 + \varepsilon^2 > 0$. The quadratic equation for C_A gives

$$C_A(1 + |\zeta_A|^2) = b_0, \quad C_A \zeta_A = A, \quad |1 - Ae^{i\theta}|^2 + \varepsilon^2 = C_A |1 - \zeta_A e^{i\theta}|^2.$$

For $u > 0$, $\rho = |\zeta_A|$ is the root in $(0,1)$ of $\rho + \rho^{-1} = u + (1+\varepsilon^2)/u$. The right side is at least $2\sqrt{1+\varepsilon^2}$, with equality at $u = \sqrt{1+\varepsilon^2}$. Since $\rho + \rho^{-1}$ decreases on $(0,1)$, the asserted maximum follows. At $u = 0$ it is immediate. Expand $\log(1-z) = -\sum_{m \ge 1} z^m/m$ for $|z| < 1$ and add the conjugate. The common bound $\rho_\varepsilon < 1$ justifies uniform convergence even as A varies. Finally $2\sum_{m \ge M} \rho_\varepsilon^m/m \le 2\rho_\varepsilon^M/[M(1-\rho_\varepsilon)]$. □

Theorem 3.2 (Sharper later-density interference)**.** *Assume the exact phase supports from (3), positive probability densities, and $k_{j+1} \geq Qk_j$ for an integer $Q \geq 4$. For every $N \geq j$, $r \in [0,1]$,*

$$\left| \int_S U_{k_j}(r\xi)(W_N - W_j)\,\mathrm{d}\sigma \right| \leq \frac{4\rho_\varepsilon^{Q-1}}{(Q-1)(1-\rho_\varepsilon)}. \tag{8}$$

Proof. The sum of prior degrees satisfies $\ell_i \leq [Q/(Q-1)]k_i$. Each increment $W_l - W_{l-1}$, $l \geq j+1$, has no phase frequency of absolute value below $k_l - \ell_{l-1}$. Moreover

$$k_l - \ell_{l-1} \geq \left(Q - \frac{Q}{Q-1}\right) k_{l-1} > (Q-2)k_j.$$

Thus $W_N - W_j$ is orthogonal to all common-phase functions of frequency mk_j, $|m| \leq Q-2$. Apply Lemma 3.1 with $A = aP_{k_j}(r\xi)$ and phase $k_j\theta$. The coefficient at positive index m transforms by $e^{imk_j\theta}$ under common phase, including its dependence on ξ, because C_A depends only on $|A|$. The phase-constant coefficient is also annihilated by orbit averaging. After discarding these frequencies the remaining test has sup norm at most E_{Q-1}. Since $\int |W_N - W_j|\,\mathrm{d}\sigma \leq 2$, (8) follows. No uniform sup norm of W_N is used. □

A sufficient replacement for the source's interference restriction is

$$Q - 1 \geq \max\left\{3,\ \frac{\max\{0, \log[8/(d_2(1-\rho_\varepsilon))]\}}{\operatorname{arsinh}\varepsilon}\right\}. \tag{9}$$

Choose an integer Q satisfying this and the other source thresholds. Then the bound is at most $d_2/2$, so the source's $d_3 = d_2/2$ is unchanged. Indeed $\rho_\varepsilon^{Q-1} \leq d_2(1-\rho_\varepsilon)/8$ when the logarithm is positive; when it is nonpositive the same conclusion follows from $d_2(1-\rho_\varepsilon) \geq 8$. The omitted factor $(Q-1)^{-1} \leq 1$ only weakens this sufficient choice.

The source instead bounds the phase coefficients by B_ε/m^2, with $B_\varepsilon = 2aC_P/\varepsilon^2 + 4a^2C_P^2/\varepsilon^4$, and uses $4B_\varepsilon/(Q-3)$. For fixed a, C_P, d_2, our sufficient interference threshold is $O(\varepsilon^{-1}\log(1/\varepsilon))$, rather than the $O(\varepsilon^{-4})$ restriction supplied by that estimate. If d_2 also varies, retain its explicit logarithm in (9); no uniform assertion about its dependence is made. The density, packing, and centered-chart thresholds can still dominate. This is a parameter-bound improvement, not a measured runtime improvement.

4 Nongeometric degrees and fixed-order weighted derivatives

Proposition 4.1 (All fixed phase orders)**.** *Let $s > 0$ be a trigonometric polynomial of degree at most $2q$, $q \geq 1$, with*

$$s(\theta + u) \leq A(1 + q|u|)^\nu s(\theta) \quad (-\pi \leq u \leq \pi).$$

For every fixed integer $d \geq 0$ there is $C_d = C_d(A,\nu)$ such that $|s^{(d)}(\theta)| \leq C_d q^d s(\theta)$.

Proof. Fix a smooth compactly supported cutoff χ equal to one on $[-2,2]$. The periodic kernel $K_q(u) = \sum_{h\in\mathbb{Z}} \chi(h/q)e^{ihu}$ reproduces s. The coefficients of $K_q^{(d)}$ are $(ih)^d\chi(h/q)$. Their Nth finite differences have sum of absolute values at most $C_{d,N}q^{d+1-N}$: express each difference as an integral of the Nth derivative of $(it)^d\chi(t/q)$ over $[0,1]^N$, whose support has length $O(q)$ and size $O(q^{d-N})$. Summation by parts, together with the elementary coefficient sum $O(q^{d+1})$, therefore gives

$$|K_q^{(d)}(u)| \leq C_{d,N}q^{d+1}(1 + q|u|)^{-N}.$$

Use $s^{(d)} = K_q^{(d)} * s$, the growth assumption, and $N > \nu + 1$. The integral of $q(1+q|u|)^{\nu-N}$ is bounded uniformly in q, proving the estimate. Constants are allowed to depend on d. □

The source proves the $d = 2$ case by this method. The proposition records the full fixed-order consequence, not a new Bernstein principle. It controls phase derivatives, not arbitrary spatial curvature derivatives.

Theorem 4.2 (Degree-sequence transfer)**.** *Use the source's peak families and profiles. Let the integer degrees satisfy $k_1 \ge k_*$ and $k_{j+1} \ge Qk_j$, where a single integer Q exceeds the source's positivity thresholds and (9). Suppose also*

$$\Gamma = \limsup_{j\to\infty} \frac{\log k_j}{j} < \infty. \tag{10}$$

Then the same corrections define positive representing densities and, for any $L > 5\Gamma/d_3$,

$$\phi = \psi + L\sum_{j\ge 1} U_{k_j}$$

has the source's centered metric-jet bounds and no holomorphic minorant $\operatorname{Re} H \le C_H + 4\log(1/(1-|z|)) + \phi$ *on B. Consequently the source's geometric transfer applies to this ϕ. No divisibility or constant-ratio condition on the degrees is needed.*

Proof. We give each transfer step. First, $\ell_{j-1} \le Qk_{j-1}/(Q-1) \le k_j/(Q-1)$, so the exact endpoint calculation and bandwidth induction still hold. The source's positivity proof only uses these inequalities and the consecutive ratios $k_{j-1}/k_j \le Q^{-1}$, as follows. If l is the last index with $k_l|u| \le 1$, the bounds (5) give the orbit estimate

$$s(\theta+u) \le e^{K_*}(2/c_0)(1+k_n|u|)^{\log_2(2/c_0)} s(\theta).$$

Indeed compare the density at level n to level l at both points; their ratio costs $(2/c_0)^{n-l}$, and phase integration at level l costs e^{K_*}. Since $k_n/k_{l+1} \ge 2^{n-l-1}$, the displayed polynomial bound follows. If no such positive l exists use $W_0 = 1$ and the same argument from level one. Proposition 4.1 with $d = 2$ gives the same C_1/Q^2 ratio used in the source. Horizontal comparison from a point to the patch center costs at most $e^{K_* D_R/\sqrt{Q}}$. All the normalized patch derivative estimates and the ordered choices c_0, C_2, K_*, C_1, Q are thus unchanged. This closes (5) by precisely the source's induction.

Second, the same-stage negative test depends on k_j only through the uniform packing and $O(k_j^{-1})$ errors; $k_1 \ge k_*$ handles these. Orthogonality uses $\ell_{j-1} < k_j$. Theorem 3.2 then gives

$$\int U_{k_j}(r\xi) W_N \,\mathrm{d}\sigma \le -d_3 \quad (j \le N,\ 1-\eta \le r^{k_j} \le 1).$$

Set $r_N = 1 - \eta/k_N$, with $0 < \eta < 1$. Bernoulli's inequality gives the displayed condition for $j \le N$. For $l = N+m$, $r_N^{k_l} \le e^{-\eta Q^m}$, so the absolute contribution of the remaining tests is bounded uniformly in N by $C\sum_{m\ge 1} e^{-\eta Q^m}$, using positivity and unit mass. Also $\psi(r_N) + 4\log(1/(1-r_N)) \le 5\log k_N + C$. It follows that the average of the proposed minorant upper bound is at most $5\log k_N - Ld_3 N + C$. This tends to $-\infty$ by (10) and the strict choice of L, whereas the average of $\operatorname{Re} H(r_N\xi)$ is the fixed number $\operatorname{Re} H(0)$. This contradiction proves the minorant assertion.

Third, on compact subballs, P_k and each fixed derivative decay geometrically in k; the series for ϕ is locally smooth. Each U_k is plurisubharmonic since

$$\partial_i\bar\partial_j U_k = \frac{a^2\varepsilon^2 P_{k,i}\overline{P_{k,j}}}{(|1-aP_k|^2+\varepsilon^2)^2},$$

so $\partial\bar\partial\phi \ge I$. In a ball-centered chart at radius r, set $\delta = 1 - r$. The source's anisotropic Cauchy proof, for any fixed positive derivative order d, gives

$$|D^d(U_k \circ T_r)(0)| \le C_{d,\varepsilon} \begin{cases} \sqrt{k\delta}, & k\delta \le 1, \\ e^{-ck\delta}, & k\delta > 1. \end{cases} \tag{11}$$

For clarity, the low-degree proof uses the polydisk radii $1/k, 1/\sqrt{k}$ about $(r, 0)$, on which the homogeneous bound gives $|P_k| \le C_P e^2$. Its first derivatives are bounded by $Ck, C\sqrt{k}$. The centered automorphism has displacements $O(\delta), O(\sqrt{\delta})$ on a fixed smaller chart polydisk. Subtracting its central value gives $O(\sqrt{k\delta})$; Cauchy estimates give every fixed positive order. For the high degrees the chart image lies a distance at least $c\delta$ from the sphere, giving the exponential bound and all its fixed derivatives by Cauchy. In the smooth composition with F_ε, every positive-order term contains a positive-order derivative of P_k; its remaining factors are bounded, giving (11). The low-degree sum is at most $(1 - Q^{-1/2})^{-1}$ and the high-degree sum is at most $\sum_{m\ge 0} e^{-cQ^m}$. In particular orders two, three, and four are uniformly bounded. The source's geometric transfer now has all its stated hypotheses. □

For example, integer degrees with successive ratios alternating between Q and $Q + 1$ satisfy the hypotheses but are not geometric. More generally only the lower ratio and the finite exponential rate are required, not a uniform upper ratio. Conversely lower separation alone permits $\log k_N/N \to \infty$; the negative contribution linear in N then need not defeat the $5 \log k_N$ boundary cost. This is a real obstruction to this proof, not a conclusion that the resulting manifold must possess bounded coordinates.

5 The exact metric and real-plane curvature

We now fix ϕ satisfying the source's centered-chart bounds $cI \le h = \partial\bar\partial\phi \le CI$, with first and mixed second component derivatives bounded uniformly in those charts. Here I denotes the identity matrix at the chart center, and its first derivatives there vanish. Keep this convention distinct from ordinary Euclidean coordinates at an arbitrary point of the ball. For (1) put

$$t = e^{\phi}|w|^2, \quad x = \frac{t}{1-t}, \quad q = x(1+x), \quad G = \lambda I + xh, \quad \theta = \mathrm{d}w + w\partial\phi.$$

The exact smooth metric is

$$\widehat{g} = \partial\bar\partial K_\lambda = G + \frac{e^{\phi}}{(1-t)^2}\theta \otimes \bar\theta, \qquad g = 2\,\mathrm{Re}\,\widehat{g}. \tag{12}$$

Away from $w = 0$, a holomorphic linear and quadratic gauge of $v = \log w$ removes the first and pure second base derivatives of $\log t$ at the point. Normalizing the vertical direction by $q^{-1/2}$ gives a rank-one vertical form V with coefficient one. The source's complete tensor identity is

$$\begin{aligned} R &= -\lambda S_I - qS_h - S_V - (S_{D+V} - S_D - S_V) + T_4 + T_3, \\ D &= (1+2x)h - qhG^{-1}h, \qquad (S_P)_{a\bar b c\bar d} = P_{a\bar b}P_{c\bar d} + P_{a\bar d}P_{c\bar b}. \end{aligned} \tag{13}$$

T_4 has four horizontal indices; T_3 exactly three. In the source's index convention,

$$\begin{aligned} (T_4)_{i\bar j k\bar l} &= -xh_{i\bar j,k\bar l} + x^2 G^{p\bar e} h_{i\bar e,k} h_{p\bar j,\bar l}, \\ (T_3)_{i\bar j k\bar 0} &= \sqrt{q}(-h_{i\bar j,k} + xG^{p\bar e} h_{i\bar e,k} h_{p\bar j}). \end{aligned}$$

Repeated indices are summed. The coefficients $G^{p\bar e}$ used in these contractions are transposed relative to the ordinary matrix inverse indexing. All symmetry-related components are included. At $w = 0$ a smooth fiber coordinate gives $G = \lambda I$, $D = h$, $T_4 = T_3 = 0$, and $R_{0\bar 0 0\bar 0} = -2$. A singular logarithmic fiber coordinate is not used to establish regularity at this section.

For the quantitative results, fix $C_{\mathrm{e}} \geq 0$ satisfying

$$\|T_4\|_F \leq C_{\mathrm{e}} x, \qquad \|T_3\|_F \leq C_{\mathrm{e}} \sqrt{q}. \tag{14}$$

The norms are the full tensor Frobenius norms in the centered horizontal and normalized vertical frame, not the maximum of a single displayed component. Finite-dimensional norm equivalence and the source's uniform jet bounds supply such a constant. For $\lambda \geq \max(1, C)$ the source's algebra gives $(1 + x)h \leq D \leq (1 + 2x)h$.

For a real plane with independent real vectors, let u_1, u_2 be their $(1, 0)$ parts and define

$$a_P = P(u_1, \bar{u}_1)P(u_2, \bar{u}_2) - \operatorname{Re} P(u_1, \bar{u}_2)^2, \quad b_P = \operatorname{Im} P(u_1, \bar{u}_2), \quad m_P = a_{P+V} - a_P - a_V.$$

The notation $\operatorname{Re} P(u_1, \bar{u}_2)^2$ here means $(\operatorname{Re}(P(u_1, \bar{u}_2)))^2$. For a Hermitian positive-semidefinite P, $a_P \geq b_P^2$; for the rank-one V, $a_V = b_V^2$. Put

$$\Xi^{a\bar{b}} = u_1^a \bar{u}_2^b - u_2^a \bar{u}_1^b, \qquad \mathcal{H}_T = T_{u_1 \bar{u}_1 u_2 \bar{u}_2} - \operatorname{Re} T_{u_1 \bar{u}_2 u_1 \bar{u}_2}.$$

The Kähler symmetries give

$$\mathcal{H}_T = -\tfrac{1}{2} T_{\bar{a}bc\bar{d}} \Xi^{a\bar{b}} \Xi^{c\bar{d}}, \qquad \mathcal{H}_{S_P} = a_P + 3b_P^2, \qquad K_g = \frac{\mathcal{H}_R}{2a_{G+V}}. \tag{15}$$

In particular the exact negative-curvature numerator is

$$\mathcal{F} = -\mathcal{H}_R = \lambda(a_I + 3b_I^2) + q(a_h + 3b_h^2) + 4a_V + m_D + 6b_D b_V - \mathcal{H}_{T_4} - \mathcal{H}_{T_3}. \tag{16}$$

The term $6b_D b_V$ has no fixed sign. Replacing it by a positive quadratic surrogate would change the problem.

6 A strengthened Gram inequality

Theorem 6.1 (Mixed-block coercivity with the full beta gap)**.** *Let $h > 0$ be a Hermitian form on the horizontal space, let V be the rank-one vertical form, and let E be horizontal with $h \leq E \leq (2 - \beta)h$, where $\beta \leq 1$ is any real number. For a real pair as above set*

$$A = a_h, \quad U = a_V, \quad J = m_h + 2b_h b_V, \quad Z = \left(\sqrt{A + 3b_h^2} - 2\sqrt{U}\right)^2.$$

Then $J \geq 0$, and if $A > 0$ and $\tau = |b_h|/\sqrt{A}$,

$$P_h(E) := A + 3b_h^2 + 4U + m_E + 6b_E b_V \;\geq\; J + Z + 2\beta(1 + \tau)\sqrt{AU}. \tag{17}$$

The formula extends continuously to $A = 0$. In an h-orthonormal horizontal frame, $J = \frac{1}{2}\|\Xi_{\mathrm{mix}}\|_h^2$, with both mixed blocks included.

Proof. The identity for J is obtained by expanding the two mixed blocks:

$$\sum_i |\Xi^{i\bar{0}}|^2 = m_h + 2b_h b_V, \qquad \sum_i |\Xi^{0\bar{i}}|^2 = m_h + 2b_h b_V.$$

This proves nonnegativity and the norm statement. Write $F = E - h$, so $0 \leq F \leq (1 - \beta)h$. For $A > 0$ a real change of basis makes the horizontal pair h-real-orthonormal; all the expressions scale

by the square of the real determinant, so we may prove the ensuing estimates in that basis. The imaginary h inner product then has absolute value $\tau \le 1$. For a positive form $0 \le F \le Hh$,

$$|b_F| \le \frac{H(1+\tau)}{2}\sqrt{A}. \tag{18}$$

Indeed the difference of the F-squared norms of $u_1^H + iu_2^H$ and $u_1^H - iu_2^H$ is, up to sign, $4b_F$. Each is nonnegative and is at most the corresponding Hh norm, which is one of $2H(1+\tau), 2H(1-\tau)$. Their absolute difference is therefore at most $2H(1+\tau)$. Rescale back to obtain (18).

The real 2 by 2 Gram matrices also give $m_F \ge 2\sqrt{a_F U} \ge 2|b_F|\sqrt{U}$. For completeness, their mixed determinant is the trace of the adjugate of one times the other. Diagonalizing a positive definite one reduces its lower bound to $r + s \ge 2\sqrt{rs}$; singular cases follow by adding a positive multiple of the identity and taking the limit. Since $m_E = m_h + m_F$ and $b_E = b_h + b_F$, subtract J to obtain

$$\begin{aligned} P_h(E) - J &= A + 3b_h^2 + 4U + m_F + 6b_F b_V + 4b_h b_V \\ &\ge A + 3b_h^2 + 4U - 4|b_F|\sqrt{U} - 4|b_h|\sqrt{U} \\ &\ge A + 3b_h^2 + 4U - 2(1+3\tau)\sqrt{AU} + 2\beta(1+\tau)\sqrt{AU}. \end{aligned}$$

Finally $1 + 3\tau \le 2\sqrt{1+3\tau^2}$, because the difference of the squares of the right and left sides is $3(1-\tau)^2$. Completing the square proves (17). At $A = 0$ or $U = 0$ the expressions extend by continuity; the product $\tau\sqrt{AU}$ is read as $|b_h|\sqrt{U}$. No sign assumption on β was used. □

The source proves the square and beta-gap terms, and proves that its zero set has $\Xi_{\mathrm{mix}} = 0$. The additional J in (17) quantifies this structure. For $h = I$, $E = 2I$, and $u_1 = (e_1, 1)$, $u_2 = (ie_1, -i)$, one has $A = U = 1$, $b_h b_V = -1$, $m_E = 4$, and $P_h(E) = 4 + 4 + 4 - 12 = 0$. Here $J = Z = 0$. Thus a strictly positive coercive bound on the whole plane cannot hold for this limiting leading expression. The mixed-block norm is precisely a quantity that may be controlled.

7 An explicit sufficient curvature parameter

Theorem 7.1 (Quantitative replacement for the sign threshold)**.** *Assume the actual tensor identity (13), Kähler symmetries, the component types of T_4, T_3, $cI \le h \le CI$, and the full Frobenius bounds (14), uniformly at all points. Define*

$$\begin{aligned} c_1 &= \frac{c}{\max(1, C)}, & D_* &= \frac{2C_{\mathrm{e}}}{c^2} + \frac{2C_{\mathrm{e}}^2}{c^3} + \frac{9}{8}, \\ a_* &= \min\{C^{-2}, 3c_1/16\}, & X &= \max\{1, (4D_*/a_*)^2\}, \\ \lambda_* &= \max\left\{1, C,\ 18C^2(1+X)^2 + (2C_{\mathrm{e}} + 4C_{\mathrm{e}}^2/c)X + 1\right\}. \end{aligned} \tag{19}$$

Then every $\lambda \ge \lambda_*$ gives $\mathcal{F} > 0$ on every nondegenerate real plane for all $x \ge 0$. These are sufficient, not optimized, constants.

Proof. First take $x \ge 1$ and rescale horizontal vector components by $q^{1/4}$; a tensor with r horizontal slots acquires $q^{-r/4}$. Evaluate all the following plane quantities on the rescaled pair. Put $E = D/\sqrt{q}$. Because h and G commute, for an eigenvalue $d \in [c, C]$,

$$\frac{E}{h} = \frac{1+2x}{\sqrt{q}} - \frac{\sqrt{q}\,d}{\lambda + xd}.$$

The lower bound $E \geq h$ follows from $D \geq (1+x)h$. Using $\sqrt{4+1/q} \leq 2 + 1/(4q) \leq 2 + 1/(4x^2)$ and $\lambda + xd \leq \max(1, C)(\lambda + x)$ gives

$$h \leq E \leq (2-\beta)h, \qquad \beta = b - e, \quad b = c_1 \frac{x}{\lambda + x} \in (0, 1], \quad e = \frac{1}{4x^2}.$$

This includes negative β, which is why that case was retained in Theorem 6.1.

We next make the tensor-norm constants explicit. The all-horizontal block has $\|\Xi_{HH}\|_F^2 = 2(a_I + b_I^2) \leq 4a_I$. Since $a_I \leq A/c^2$ and $\|\Xi_{\mathrm{mix}}\|_F^2 \leq 2J/c$, contraction (15) yields

$$\begin{aligned} |\mathcal{H}_{\widetilde{T}_4}| &\leq \frac{2C_{\mathrm{e}}}{c^2 x} A, \\ |\mathcal{H}_{\widetilde{T}_3}| &\leq \frac{C_{\mathrm{e}} q^{-1/4}}{\sqrt{2}} \|\Xi_{HH}\|_F \, \|\Xi_{\mathrm{mix}}\|_F \leq \frac{2C_{\mathrm{e}}}{c^{3/2}\sqrt{x}} \sqrt{AJ} \leq \frac{J}{2} + \frac{2C_{\mathrm{e}}^2}{c^3 x} A. \end{aligned} \tag{20}$$

The factor $1/\sqrt{2}$ in the first T_3 bound follows by Cauchy–Schwarz on the two disjoint tensor blocks $HH \otimes \mathrm{mix}$ and $\mathrm{mix} \otimes HH$ in the contraction, including the prefactor $1/2$. The full Frobenius norm includes both blocks. This reasoning would fail for an unstructured error tensor.

Let $s_\tau = \sqrt{1 + 3\tau^2} \in [1, 2]$. Since $\sqrt{U} \leq \sqrt{A} + \sqrt{Z}/2$,

$$4e\sqrt{AU} \leq 4eA + 2e\sqrt{AZ} \leq Z/2 + (4e + 2e^2)A.$$

The negative part of the beta term in (17) is at most $4e\sqrt{AU}$. The positive part is at least $2b\sqrt{AU}$. Also

$$\tfrac{1}{2} Z + b\sqrt{AU} \geq \frac{3b}{16}(A + U). \tag{21}$$

To prove this, in variables $\sqrt{A}, \sqrt{U}$ the left side has symmetric matrix

$$\begin{pmatrix} s_\tau^2/2 & (b - 2s_\tau)/2 \\ (b - 2s_\tau)/2 & 2 \end{pmatrix}.$$

Its determinant is $bs_\tau - b^2/4 \geq 3b/4$ and its trace is at most 4. The diagonal is positive, so its smallest eigenvalue is at least determinant divided by trace.

The rescaled version of (16) is $\lambda(a_I + 3b_I^2)/q + P_h(E)$ minus the two errors. Use (17), (20), (21), $A \leq C^2 a_I$, and $4e + 2e^2 \leq 9/(8x)$ to conclude

$$\mathcal{F} \geq \left[\frac{\lambda}{C^2 q} + \frac{3b}{16} - \frac{D_*}{x} \right] A + \frac{3b}{16} U + \frac{J}{2}. \tag{22}$$

An additional positive $b\sqrt{AU}$ has simply been dropped. For $x \geq 1$ and any $\lambda > 0$,

$$\frac{\lambda}{q} + \frac{x}{\lambda + x} \geq \frac{1}{2\sqrt{x}}.$$

If $\lambda \geq x^{3/2}$ the first term suffices since $q \leq 2x^2$; otherwise the second suffices. Thus, for $x \geq X$, the coefficient of A in (22) is at least $a_*/(2\sqrt{x}) - D_*/x \geq a_*/(4\sqrt{x}) > 0$. The coefficients of U, J are also positive. They cannot all have zero arguments on a nondegenerate plane: if $A = U = J = 0$, then $b_h = b_V = 0$, $m_h = 0$, and $a_{h+V} = 0$, contradicting positive definiteness of $h + V$.

It remains to cover $0 \leq x \leq X$ without rescaling. Here $m_D \geq c(1+x)m_I$ and $|b_D| \leq 2C(1+x)\sqrt{a_I}$. Consequently

$$|6 b_D b_V| \leq 2U + 18C^2(1+x)^2 a_I.$$

The same Frobenius contraction estimates, now using $\|\Xi_{\mathrm{mix}}\|_F^2 \le 4m_I$, give

$$|\mathcal{H}_{T_4}| \le 2C_{\mathrm{e}} x a_I, \qquad |\mathcal{H}_{T_3}| \le 2\sqrt{2}C_{\mathrm{e}}\sqrt{q}\sqrt{a_I m_I} \le \tfrac{c}{2}(1+x)m_I + \frac{4C_{\mathrm{e}}^2}{c} x a_I.$$

Retaining only nonnegative remaining terms in (16) yields

$$\begin{aligned}\mathcal{F} \ge [\lambda - 18C^2(1+x)^2 - (2C_{\mathrm{e}} + 4C_{\mathrm{e}}^2/c)x]a_I \\ + 2U + \tfrac{c}{2}(1+x)m_I.\end{aligned} \tag{23}$$

The first coefficient is at least one by (19). On a nondegenerate plane not all a_I, U, m_I vanish, since $a_{I+V} = a_I + U + m_I > 0$. This proves strict positivity, including the zero section $x = 0$ in its smooth frame. □

This is a source-relative improvement at the uniform sign step. The source's proof obtains a finite threshold by compactness: at a limiting zero, $\Xi_{\mathrm{mix}} \to 0$ improves the evaluated three-horizontal error from $O(x^{-1/2})$ to $o(x^{-1/2})$. The positive gap $\lambda/q + x/(\lambda + x) \ge 1/(2\sqrt{x})$ then dominates. Here (20) leaves a quantitatively bounded $O(A/x)$ error instead. We have not inferred a numerical value of C_{e} from finitely many sampled jets, and thus do not report a numerical metric parameter for the infinite construction.

8 What the local results do and do not imply globally

For each fixed finite $\lambda \ge \lambda_*$, the source's end calculation is

$$\widetilde{g}_x = G/\sqrt{q} + V = h + V + O_\lambda(x^{-1}), \qquad \widetilde{R}_x = -S_{h+V} + O_\lambda(x^{-1/2}). \tag{24}$$

Indeed $D = xh + \lambda(1+x)hG^{-1}$, so $D/\sqrt{q} = h + O_\lambda(x^{-1})$. Four-horizontal errors rescale to $O(x^{-1})$ and three-horizontal errors to $O(x^{-1/2})$. The limiting quotient is

$$-\frac{1}{2}\left(1 + 3\,\frac{b_{h+V}^2}{a_{h+V}}\right) \in [-2, -1/2].$$

Normalize real pairs with respect to $I + V$. Then $a_{h+V} \ge \min(c, 1)^2$, making this convergence uniform over all real planes and permitted data. In particular, for a sufficiently large end cutoff depending on the fixed λ, $-3 \le K_g \le -1/4$. On the bounded x range, the allowed matrices, tensor errors, and normalized pairs form a compact set. Theorem 7.1 is strict on this whole set; the denominator is uniformly positive. Compactness gives both curvature bounds there. This uses *fixed* λ; (24) alone is not uniform as $\lambda \to \infty$.

Completeness is a different assertion. The map $(z, w) \mapsto (z, e^{\phi(z)/2}w)$ is a smooth real diffeomorphism to $B \times \mathbb{B}^1$, hence gives contractibility, not a holomorphic trivialization. From the exact metric,

$$|\nabla_g \psi|^2 \le 2/\lambda, \qquad |\nabla_g[-\log(1-t)]|^2 = 2t \le 2.$$

Bounded metric balls therefore keep the base inside a compact subball and t away from one. On that compact base ϕ is bounded, so the fiber also remains in a compact region of the domain. A geodesic of finite length cannot escape such a set; smoothness and the ordinary geodesic ODE extend it. Small local tensor residuals without these barriers would not establish completeness.

Finally suppose $F : M_\phi \to \mathbb{C}^3$ is bounded and has nonvanishing Jacobian everywhere. On the zero section let $d(z) = \det DF(z, 0)$. The base Cauchy bounds are $O((1 - |z|)^{-1})$ and the fiber bound is $O(e^{\phi/2})$. Expanding the three by three determinant gives

$$\log |d(z)|^2 \le C_F + 4 \log \frac{1}{1 - |z|} + \phi(z).$$

The nonzero holomorphic function d has a holomorphic logarithm on the simply connected ball. Taking $H = 2 \log d$ contradicts the minorant exclusion in Theorem 4.2 (or in the original geometric-degree source). The density averages and infinite negative drift are indispensable to this step; curvature pinching alone is not substituted for it.

9 Ram reformulation and actual private proof transfer

The relevant Ram object here is a constrained positive averaging functional with exact phase nullspace, followed by derivative-weighted geometric estimates. It is not an arbitrary matrix inverse. The original local-moments chapter [5], lines 1–380, was read directly, including its Taylor proof, degree-closure proof, full product rule, support-tail discussion, and realizability warning. Included files referenced by its input commands were not thereby read. For an output-centered kernel it proves, under integrable absolute moments and derivative control on all contributing segments,

$$Tf(x) = \sum_{|\alpha| \le n} \frac{(-1)^{|\alpha|} m_\alpha(x)}{\alpha!} D^\alpha f(x) + r_n(x), \qquad |r_n| \le \frac{M_{n+1}}{(n+1)!} \sup \left\| D^{n+1} f \right\|.$$

Its proof is Taylor's formula on $t \mapsto f(x - tu)$ followed by integration. Positivity is a property of the original measure, not of its differential truncation. In M688 the endpoint identity (4), rather than finite Taylor truncation, is the stronger exact annihilation mechanism.

The actual jet module [6], lines 513–864, was read through the complete full-Leibniz, endpoint, and differentiated-remainder proofs. Its correct differentiated row is

$$D^k (S_n f^{(n)}) = \sum_{p=0}^{k} \binom{k}{p} S_n^{(k-p)} f^{(n+p)}.$$

Coefficient derivatives do not disappear merely because $n + k$ exceeds a truncation degree. With moving integration limits, the endpoint derivative acts in both slots. Its remainder proof separates the differentiated integral remainder from omitted higher field derivatives. For M688 these observations have concrete force: the s'_p term in (3), the two endpoint values in (4), and the mixed derivatives $h_{i\bar{j},k\bar{l}}$ in (13) cannot be discarded. The chart estimates (11) supply derivative bounds directly; a small scalar phase tail alone would not supply them.

The actual core-pat-16 report [7], lines 209–350, was read, including the covariance, Gaussian-quadratic, and remainder proofs. For features square integrable under one probability law it proves

$$M = \mathbb{E}[(\Phi - \mathbb{E}\Phi)(\Phi - \mathbb{E}\Phi)^*] \succeq 0.$$

Its proof is the identity $v^* M v = \mathbb{E}|v^*(\Phi - \mathbb{E}\Phi)|^2$. The same proof works for finite phase features under $W_j \, d\sigma$. The direction is from an actual law to a positive matrix, not the converse. Its example with mean zero and second moment two is positive as a moment matrix but unrealizable on $[-1, 1]$. Thus finite PSD checks cannot replace construction of the positive W_j, their support, or all-degree annihilation. The report's covariance error bound $2\sqrt{V_G}\epsilon + \epsilon^2$ assumes an actual common L^2 remainder. It is not a pointwise curvature bound and is not used as one here.

The private passages supply disciplined handling of moments, derivatives, and remainders. The improvements in Theorems 3.2, 4.2, 6.1, and 7.1 additionally require the particular source's phase spectrum, lacunarity, and tensor-slot geometry. They do not follow from relabeling a finite linear system. The strengthened Gram route and its threshold were developed during the batch mathematical review and independently checked in the derivations above; that collaboration is not a claim of external peer review or priority.

10 Current primary literature and source status

The public refresh on October 8, 2026 used only public titles and arXiv identifiers. Wu and Yau [2], Theorems 2, 3, 6 and Corollary 7, distinguish holomorphic pinching from real sectional pinching and the additional simply-connected hypothesis in the Bergman comparison. Their equivalence of invariant metrics is not a theorem providing one global bounded holomorphic coordinate map. It therefore is not a replacement for either the minorant obstruction or the real-plane calculation here.

Hao, Chen, Wang, and Zhang [3] have a September 26, 2026 version 2, later than the version 1 cited by the pinned source. Its Theorem 2 treats a complete Kähler–Einstein disk-bundle metric under specified base Ricci and sectional-curvature conditions; its proof retains the Riemannian sectional tensor. The base and metric hypotheses differ from the variable-density M688 construction. Their Hartogs Bergman–Einstein characterization also has a simply-connected bounded base and fiber dimension at least two in Theorem 4; it is not a characterization for the one-dimensional fibers in (1). The earlier disc-bundle paper [4] is related background, not a proof of the present density construction.

The source status is pinned local mathematical text. The source-identity sidecar records byte comparisons against the captured October 8 Git trees; this is not a fresh live-branch fetch, publication validation, or theorem-validity certificate. A live GitHub directory opening did not return usable content in this session. The current arXiv refresh above is separate from that dated source-byte identity.

11 Benefits, costs, and further work

Benefits. Exact spectral factorization replaces an algebraic coefficient estimate by geometric decay uniformly in the peak amplitude. It needs only probability normalization to control later densities, avoiding any growth of their sup norms. The degree transfer admits irregular integer sequences while identifying the separate exponential-growth cost. The mixed Gram term quantitatively controls exactly the error block that becomes delicate near the oppositely oriented limiting plane. Its Young estimate yields an explicit sufficient λ_* from declared geometric bounds. The proofs cover all real planes, not a finite sampling of holomorphic directions.

Costs and boundaries. The regularization $\varepsilon > 0$ is essential: as it tends to zero, the uniform spectral radius tends to one. Density positivity still requires the source's ordered constants and may dominate the new interference threshold. Increasing the potential amplitude L changes the centered jet constants and hence c, C, C_{e}. Formula (19) can be extremely large and does not establish an optimized metric. Uniform phase derivative bounds do not by themselves control horizontal derivatives. Infinite density construction, global smoothness, completeness, and the holomorphic logarithm argument remain separate obligations. Finite exact arithmetic tests verify fixtures, not these universal or infinite assertions.

Concrete next steps. One can optimize the exact Fourier remainder using the actual $|aP_k|$ range instead of its universal maximum; this requires tracking the peak values on and off patches and cannot simply assume they avoid the maximizing radius. One can optimize λ_* by retaining Z and the discarded $b\sqrt{AU}$ in the proof and by using the actual T_3 component contractions instead of their full Frobenius envelope. A useful quantitative implementation would first derive analytic bounds for C_{e} from the packing and chart constants, then propagate them through (19); sampling finitely many points cannot supply those uniform bounds.

For degree sequences with $\log k_N/N$ unbounded, the present constant-amplitude drift does not dominate the boundary allowance. Variable amplitudes are a possible next question, but they also enter all centered derivative sums and can destroy uniform curvature input bounds. Relaxing exact phase moments to approximate ones would require error control for every holomorphic minorant under consideration, whose size and growth are not uniformly bounded; a finite moment tolerance is insufficient. Finally, extending the geometric conclusion to other base dimensions needs new packing, density, and determinant growth bookkeeping. The Gram inequality alone supplies no such global extension.

12 Evidence and contribution assessment

The literal source-read scope is all twelve build text files, 2,957 lines, including bibliography and main file. The private scope is the three bounded intervals stated in Section 9: 874 distinct lines, not three complete private works. The read ledger records exact paths, line intervals, and pre-read and final hashes. Equal endpoint hashes do not prove absence of an intervening change throughout the read. One truncated combined tool display was followed by a targeted reread of the affected local-moments tail; no unseen text is credited.

The accompanying check script exercises exact endpoint cancellation, spectral factorization, phase gaps for irregular sequences, Gram and mixed-block identities, beta coercivity, and the threshold inequalities. Its result file is bound to the final TeX hash. The batch-level check record contains additional independent algebra fixtures, not all specific to M688. Both kinds of finite checks are supporting diagnostics; the continuum claims rest on the written proofs and their stated hypotheses. PDF rendering and all-page visual inspection are recorded separately and carry no mathematical guarantee.

The realized assessment is C3/5. The meaningful improvements are the explicit spectral interference restriction and the strengthened Gram estimate with its explicit uniform sufficient curvature parameter. The degree-sequence result is a conditional transfer through the source's genuine density and chart construction. The source's exact all-degree annihilation, second-order weighted Bernstein argument, positivity induction, complete metric, and global coordinate obstruction are not counted as new contributions. There is no claim of historical novelty, speedup, exhaustive proof certification, or finite global test.

References

[1] OpenAI, *A negatively pinched Kähler threefold without bounded holomorphic coordinates*, September 25, 2026, pinned twelve-file build. Public source directory. Sections 01–09, including both 04 modules; exact local identity in sidecar.

[2] D. Wu and S.-T. Yau, *Invariant metrics on negatively pinched complete Kähler manifolds*, J. Amer. Math. Soc. 33 (2020), 103–133. arXiv:1711.09475. Primary theorem statements and Corollary 7 inspected; not a full reread.

[3] Y. Hao, M. Chen, A. Wang, and B. Zhang, *The curvature estimation of the complete Kähler–Einstein metrics on the disk bundles*, September 26, 2026. arXiv:2609.14482v2. Introduction, sectional-curvature Theorem 2 and its complete proof, and Theorem 4 statement inspected; remainder not fully audited.

[4] Y. Hao, M. Chen, and A. Wang, *On the negatively pinched properties of the disc bundles over negatively pinched Kähler manifolds*, September 4, 2025. arXiv:2509.04179v1. Abstract and metadata checked only.

[5] M. Subbarao, *Local Moments and Ram Differential Operators*, RamCalBooks, Volume 1 revision 2, chapter 3, confidential original. File `ch03-local-moments-rev2.tex`, lines 1–380 of 446.

[6] M. Subbarao, RamNonlinearIE-PDE, TR1 revision 2, jet module, confidential original. File `sec-03-jets.tex`, lines 513–864 of 1,885.

[7] M. Subbarao, *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16, confidential original. File `TR-core-pat-16.tex`, lines 209–350 of 939.